

Explainability for Traffic Sign Recognition

Ethics and Explainability of Artificial Intelligence

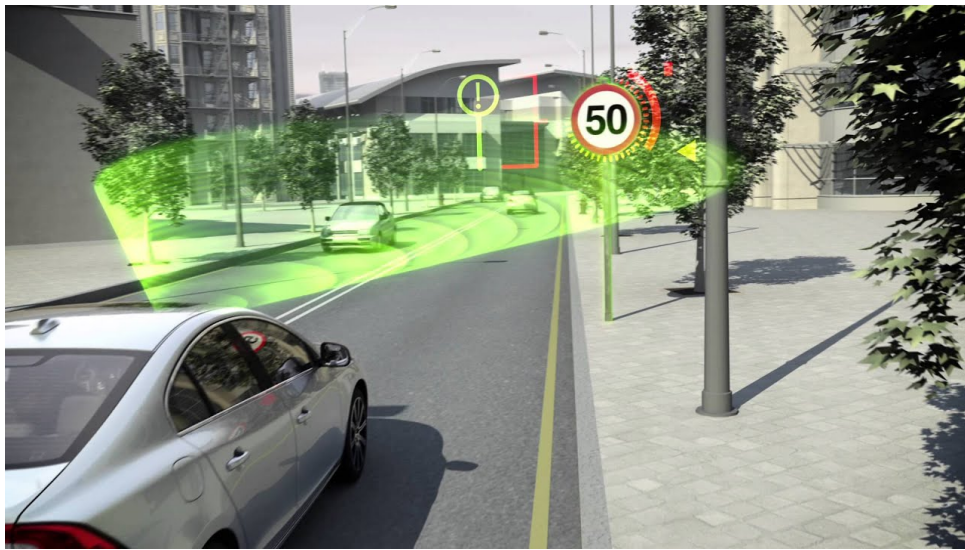


Figure 1: Traffic Sign Recognition [1]

Manuel Rodríguez Villegas

Madrid, December 31, 2025

Contents

1	Introduction	3
2	Data and Methods	3
2.1	Dataset	3
2.2	Models	3
2.2.1	Logistic Regression over HOG features	4
2.2.2	Shallow CNN	4
2.2.3	MobileNet-V2	4
2.2.4	ResNet-18	4
2.3	XAI Methods	5
2.3.1	Grad-CAM	5
2.3.2	Integrated Gradients	5
2.3.3	Occlusion Sensitivity	5
2.4	Training	6
2.4.1	Logistic Regression	6
2.4.2	Shallow CNN	6
2.4.3	MobileNet-V2	6
2.4.4	ResNet-18	6
3	Results	6
3.1	Logistic Regression Feature Importance	7
3.2	Local Explanations	7
3.3	Global Explanations	9
4	Actions and Insights	9
5	Discussion	10
	Appendix A Training Plots	11
	Appendix B Further Explanations	13

1 Introduction

This project analyzes the explainability of four computer vision models used for Traffic Sign Recognition (TSR). TSR is used in modern vehicles to recognize road signs and adjust their behavior based on that information. It enables functionalities like lane-keeping assistance, speed control, and, most importantly, autonomous driving. Although these features still sound futuristic to many, the first two are present in almost all new vehicles, as car manufacturers are required by law to integrate Advanced Driver Assistance Systems (ADAS). While full self-driving is still in a premature phase, large fleets of autonomous vehicles are already being deployed in some American cities, such as Los Angeles or San Francisco.

All these advancements will bring improved safety for humans, but it is first necessary to ensure that they perform robustly as intended. That is where AI explainability comes in: going one step beyond plain model training and evaluation to understand the model’s decision-making process. This is crucial for interpreting local explanations and general behavior, while being fundamental for understanding what the model is doing wrong and why.

TSR is a good case study for explainability because it requires a deep understanding of the model: car manufacturers need to know why their models are recognizing certain signs that trigger specific actions. These actions may involve sudden safety maneuvers, such as braking or turning, which have a direct impact on human safety. For that reason, this is a critical application of computer vision where model transparency is a priority.

TSR presents intrinsic challenges that make this task non-trivial. The main difficulty is the varying lighting conditions of the environment, which forces models to be robust in all types of situations to provide reliable performance in real use cases. This includes sunny days, rain, and low-light conditions. Secondly, different camera placements in distinct vehicles may reduce performance, as the viewing angle changes and incoming images present a positional bias. Another issue relates to the computational power of today’s vehicles, which is limited compared to workstation devices. Furthermore, recognition must be fast so the vehicle has sufficient time to react smoothly to new information. Finally, road signs can be partially tilted, covered, or painted, further increasing the difficulty of recognition.

All the code used in this project can be accessed in <https://github.com/ICAI-IMAT-XAI/final-project-rdgzmanuel>.

2 Data and Methods

2.1 Dataset

The dataset used in this project is the German Traffic Sign Recognition Benchmark (GTSRB), publicly available in Kaggle [1]. The dataset contains 51839 labeled images of road signs, with 43 classes representing the type of sign (Speed limit (20km/h), No passing, Stop, Yield, etc.). The data card can be consulted at the dataset Kaggle website. Images are usually very small, containing the correspondent sign and some background around it. This reduced size makes model training more feasible while maintaining enough information for a good performance. Data can be downloaded following the instructions of the repository README.md.

2.2 Models

TSR is a computer vision task, so the models used are predominantly Convolutional Neural Networks (CNNs). This project uses two baseline models and two more complex ones, with the aim of establishing a baseline performance to justify complexity and compare the explainability of simpler and more complex models. In particular, the baselines are a Logistic Regression classifier for the HOG features

of images and a shallow CNN with just two layers. The complex alternatives are both pretrained models that have been fine-tuned for this specific task: MobileNet-V2 [3] and ResNet-18 [2].

2.2.1 Logistic Regression over HOG features

The first model is a Logistic Regression classifier that utilizes Histogram of Oriented Gradients (HOG) features as input to output class probabilities. As a discriminative probabilistic model, it directly maps input features to output categories. This model is included in this comparative study as a transparent baseline capable of providing significant performance relative to more complex black-box architectures.

HOG is an image representation technique that combines gradient magnitude and orientation into a single vector. It serves as a lightweight descriptor that captures essential structural information, which is particularly effective for TSR since signs are largely distinguishable by their shapes and edges.

The Logistic Regression model processes these HOG vectors to provide both discrete predictions and class-wise probability distributions. As a white-box model, its internal mechanics are mathematically transparent, offering direct interpretability of the feature importance.

2.2.2 Shallow CNN

The second baseline is a shallow CNN containing only two convolutional layers followed by a classification head. To ensure stable convergence, the convolutional layers are initialized using Kaiming (He) initialization, while the linear layers utilize Xavier (Glorot) initialization. This model serves as a control baseline to contrast against deeper pretrained architectures, as it has fewer parameters and begins with no prior knowledge. Unlike the more complex CNNs used in this study, this network is trained entirely from scratch specifically for the TSR task.

2.2.3 MobileNet-V2

MobileNet-V2 is a vision architecture introduced by Sandler et al. [3] specifically designed for mobile and embedded systems with constrained computational resources. This makes it directly applicable to the automotive industry, where onboard computing devices often lack the power of high-end workstations.

MobileNet-V2 introduces the inverted residual block with linear bottlenecks, a structure that significantly optimizes the efficiency of convolutions. By expanding the input into a higher-dimensional space through a pointwise convolution, applying a 3×3 convolution to filter features, and then projecting the data back to a lower-dimensional bottleneck with a linear activation, the model minimizes information loss while maintaining a low parameter count. This architecture uses shortcut connections between bottlenecks to facilitate gradient flow, ensuring robust learning despite its compact size.

Despite its superior predictive power, the model size is still comparable to simpler baselines, making it an ideal candidate for real-time inference in ADAS. Unlike the shallow CNN, which may suffer from the vanishing gradient problem or lack representative capacity, MobileNet-V2 provides a better balance between latency and precision.

2.2.4 ResNet-18

ResNet-18 is a deep residual network introduced by He et al. [2] that revolutionized the training of deep architectures through the implementation of skip connections. These shortcuts allow the gradient to flow through the network without being excessively reduced by the repeated non-linear transformations of successive layers. By learning residual mappings instead of direct mappings, the architecture effectively mitigates the vanishing gradient problem, enabling the training of deeper networks that achieve significantly higher accuracy than shallower variants.

The increased complexity makes it particularly robust against the spatial and lighting variances inherent in the GTSRB dataset, though it presents a more significant challenge for explainability due to its depth and the non-linear interactions between its 18 layers.

The model used here is pretrained on ImageNet and subsequently fine-tuned for the TSR task. This transfer learning approach allows the model to leverage general visual patterns, such as curves and textures learned from natural images, and adapt them to the specific structures of road signs. While ResNet-18 is computationally more demanding than MobileNet-V2, it provides a performance upper bound against which the interpretability and efficiency of the other models can be measured.

2.3 XAI Methods

Three methods have been used in this project to understand the model’s internal behavior and try to explain its predictions: GradCAM, Integrated Gradients and Occlusion. The first two of them are model-specific methods, as they make use of the model’s internal gradients to compute explanations. The Occlusion method will be used to evaluate the importance of the output of the other methods, expecting that occluding important parts of the images significantly reduces the model performance.

2.3.1 Grad-CAM

Grad-CAM is a model-specific explainability technique that uses the feature representations in the final convolutional layer to generate a heatmap over the original image. Each channel is weighted by its gradient with respect to a target class, which can be the ground truth or a predicted label. This technique builds upon CAM (Class Activation Mapping) by removing the requirement for a Global Average Pooling layer, though it still requires a CNN architecture. By visualizing a gradient-weighted heatmap, this technique highlights the regions of the image that have the highest influence on the prediction when subjected to infinitesimal changes.

In this project, Grad-CAM is employed for both global and local explanations. By averaging heatmaps over all instances of a specific category, we can identify where the model tends to focus globally for that class. On the contrary, analyzing a single instance provides a local explanation for a specific prediction.

2.3.2 Integrated Gradients

Integrated Gradients is another model-specific technique that produces an attribution map of the same dimensions as the original input. This representation is constructed by aggregating the gradients along a path from a baseline (a black image in this case) to the original input. Gradients are computed with respect to a target class to determine feature importance.

Integrated Gradients satisfy the axioms of sensitivity and implementation invariance, and they are particularly effective at reducing gradient saturation artifacts that often appear in other gradient-based methods. While a primary drawback is that the explanation depends heavily on the choice of baseline, it remains one of the most widely adopted techniques for computer vision explainability.

2.3.3 Occlusion Sensitivity

Occlusion sensitivity is used as an evaluation technique for the previously mentioned XAI methods. The objective is to verify the regions of the input image that exert the greatest influence on the predicted class. By sliding an occluding patch across the original image and monitoring the classification pipeline, we can observe how the model’s predictions vary according to the patch’s position.

The underlying principle is that performance should suffer a significant drop when the patch covers the most influential parts of the image, while remaining relatively stable when less important areas are

covered. This provides a primarily visual evaluation that helps analyze the fidelity of the Grad-CAM and Integrated Gradients explanations.

2.4 Training

This section details the training procedures for the four models. The data was divided according to the original dataset structure, which provides an 85% training and 15% test split. The training set was further subdivided to include a validation set, resulting in a final distribution of 70% training (36,287 samples), 15% validation (7,776 samples), and 15% test (7,776 samples). Training was conducted using an NVIDIA GeForce RTX 4060 Laptop GPU. Training curves are shown in Appendix A and metrics are shown in Table 1.

2.4.1 Logistic Regression

The Logistic Regression model was trained using the *scikit-learn* library. Due to the high dimensionality of the HOG features, the inherent interpretability of this model, initially its main advantage, was significantly reduced. Efforts to mitigate this by reducing the HOG vector length resulted in an excessively slow training process, leaving dimensionality reduction unfeasible for this project. Consequently, the utility of this model for the explainability comparative is reduced, serving primarily as a performance baseline.

2.4.2 Shallow CNN

The training of the shallow CNN baseline was considerably more efficient, showing stable convergence and natural learning curves for both loss and accuracy. The model was trained for 90 epochs, with the entire process taking approximately 20 minutes.

2.4.3 MobileNet-V2

MobileNet-V2 was implemented as a pretrained model and subsequently fine-tuned on the GTSRB dataset. This approach allows the architecture to retain learned visual patterns while adapting to the specificities of traffic sign classification. For a stable transition, the fine-tuning was executed in two phases: the first phase unfroze only the classification head for 20 epochs with a high learning rate (1×10^{-3}). While initially unstable, this allowed the head to align with the new feature space. In the second phase, the learning rate was reduced (1×10^{-4}) for an additional 60 epochs, and the last two blocks were unfrozen to allow for deeper weight adjustment. Total training took around 20 minutes as well.

2.4.4 ResNet-18

The training of ResNet-18 followed a similar two-phase fine-tuning strategy to optimize performance. Given that this model contains a larger number of parameters compared to MobileNet-V2, only the final residual block was unfrozen during the second phase to prevent overfitting.

Looking at the results, it is evident that MobileNet-V2 underperforms significantly compared to the other models, which achieve similar scores. Analyzing the training curves, we can see that while this model maintains a margin for improvement, it requires considerably more time than the alternatives to converge. When compared to ResNet-18, the residual connections and higher complexity of the latter have a substantial impact on training stability and convergence speed.

3 Results

This section details the explainability experiments and results obtained after training the models. A separate Python notebook has been used for convenient testing. The experiments carried out include a feature importance analysis of the Logistic Regression and different XAI visualizations for the CNNs models.

Model	Accuracy	F1 Macro	F1 Weighted
Logistic Regression	0.973	0.981	0.973
Shallow CNN	0.949	0.938	0.948
MobileNet-V2	0.885	0.884	0.886
ResNet-18	0.983	0.983	0.983

Table 1: Performance comparison across the four models on the test set

3.1 Logistic Regression Feature Importance

As previously noted, the Logistic Regression model contains an excessively large feature space, which significantly limits its interpretability. An examination of the feature importance distribution reveals that weights are very dispersed, with no single feature exhibiting a dominant positive or negative influence. This analysis could be further improved by mapping the orientations and gradient magnitudes of the most significant HOG features back onto the original image to visualize the specific spatial structures affecting the classification.

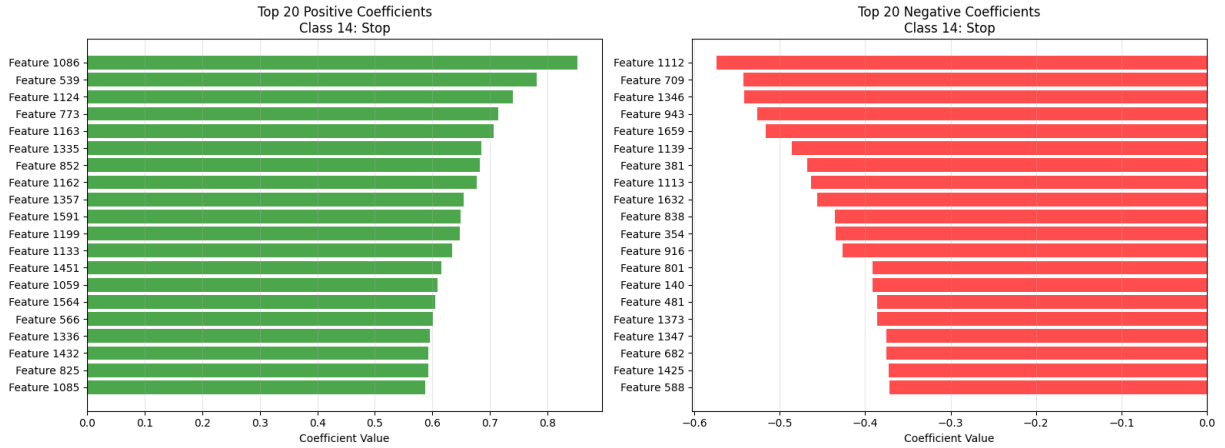


Figure 2: Logistic Regression Feature Importance.

3.2 Local Explanations

Local explanations aim to interpret the model’s decision for particular instances. They are especially relevant when stakeholders need to understand the reasoning behind specific classifications or determine which feature values could be altered to achieve a different result. This section analyzes model behavior over two specific instances: a STOP sign and an 80 km/h speed limit sign. Each instance is evaluated across every CNN architecture using the selected XAI methods.

Starting with the Shallow CNN, an interesting result emerges: both Grad-CAM and Integrated Gradients (IG) suggest the model focuses specifically on edges, particularly the top border. The Grad-CAM heatmap appears slightly misaligned with the original image, likely due to the large spatial dimensions of the feature representations in the second convolutional layer. This discrepancy should diminish in deeper networks. Occlusion Sensitivity shows almost no highlighted areas here; the low channel count and high spatial resolution of the final layers make it difficult for the occlusion mechanism to aggregate meaningful values. However, the speed limit sign presents different results, even highlighting background elements. In this case, Occlusion works effectively, focusing on the core of the sign and indicating that the other two methods may not faithfully reflect the most influential regions.

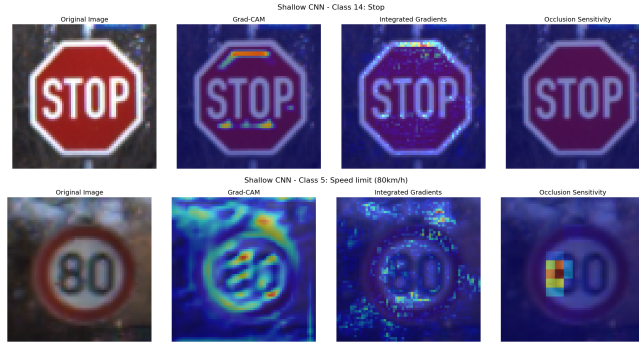


Figure 3: Local explanations for shallow CNN.

For the MobileNet-V2 model, the three methods provide more coherent representations. Grad-CAM correctly captures the region where gradients exert the highest influence, corresponding to the center of the sign. While IG also highlights edges, it places greater emphasis on the sign’s characters. This is confirmed by Occlusion Sensitivity, which shows that patching the characters produces the most significant drop in confidence. Interestingly, Grad-CAM assigns importance to the lower part of the sign, possibly referencing the post. Regarding the speed limit sign, Grad-CAM shows a highly dispersed representation reinforced by both IG and Occlusion. The model appears to focus excessively on the left background, which likely explains its inferior performance compared to the other architectures.

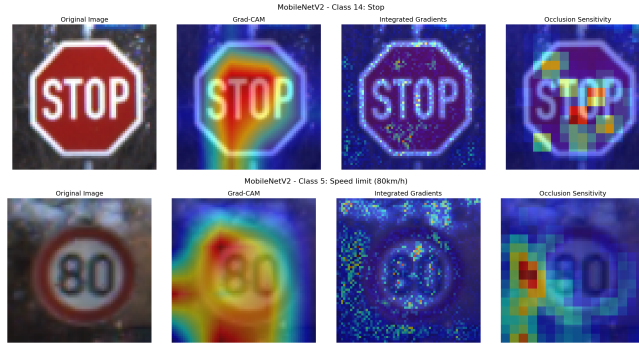


Figure 4: Local explanations for MobileNet-V2.

Finally, in the STOP sign ResNet-18 yields smoother visualizations where Grad-CAM and IG are concentrated on the sign’s characters. Occlusion Sensitivity confirms this focus while also highlighting the importance of the top edge. For the speed limit sign, Grad-CAM provides a natural representation, but IG again emphasizes the left background. Occlusion Sensitivity reinforces the importance of the central features, suggesting that IG may be capturing noise in this specific instance.

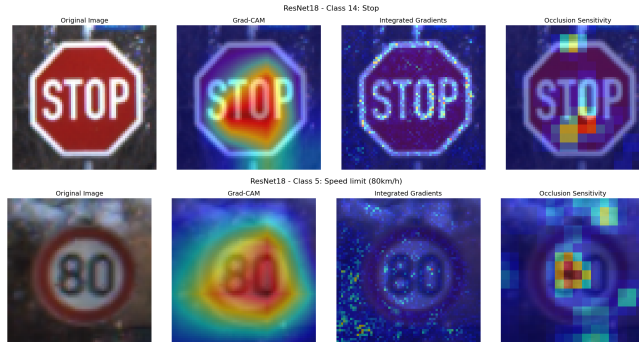


Figure 5: Local explanations for ResNet-18.

3.3 Global Explanations

Global explanations assist in understanding a model’s internal logic by aggregating results across many instances of the same category. This process effectively reduces the noise associated with individual examples and highlights the regions where the model consistently focuses. In this project, global explanations are generated by averaging Grad-CAM heatmaps over 100 instances per class.

The visualizations below confirm that the shallow CNN, despite its reduced complexity, produces highly interpretable representations while maintaining competitive performance. It is evident that its gradients are significantly more localized compared to those of the more complex models. This behavior is expected, as gradient flow from deeper architectures typically distributes over a larger area of the original input image. More examples of this case can be found in Appendix B .

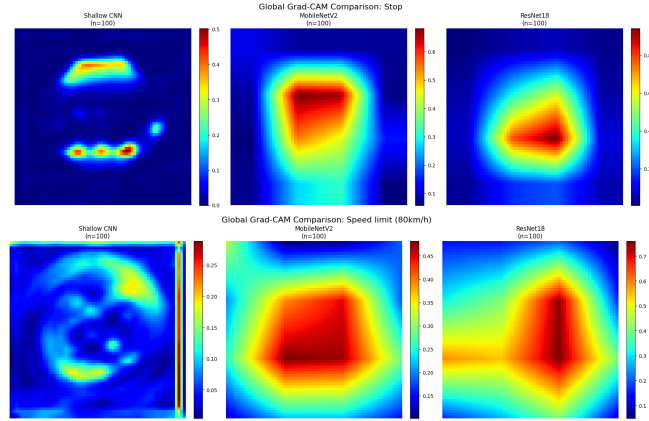


Figure 6: Global explanations.

4 Actions and Insights

The previous visualizations confirm that model failure is often caused by the model focusing on background elements rather than the sign itself. This behavior is particularly prominent in the IG and Grad-CAM representations. To mitigate this, data augmentation strategies could be implemented to decorrelate the background from the target class, such as background randomization, edge cropping, or aggressive lighting transformations. This way, the model could learn that many different background correspond to the same class if the road sign is still the same.

Another insight relates to the specific layer chosen for backpropagation in gradient-based methods. Experiments indicate that while deeper layers generally produce more semantic visualizations, exceeding a certain depth threshold can make the explanations unusable. Figure 7 illustrates this case, where excessively deep layers degrade the quality of Grad-CAM representations. IG visualizations are partially unaffected by this because of the integrating property that helps avoid gradient saturation.

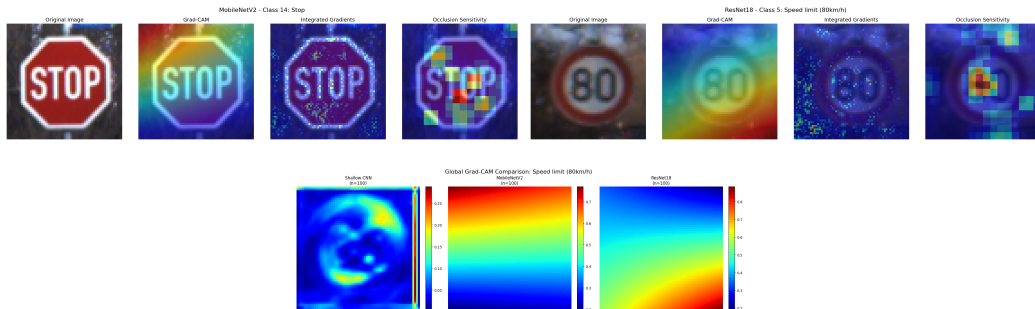


Figure 7: Explanations with backward from deeper layers.

Regarding overall performance, latency, and interpretability, MobileNet-V2 is notably behind the other candidates. Although extended training would likely improve its metrics, it would still struggle to match the performance of the other three models. Its architectural complexity does not translate into superior results for this task compared to simpler alternatives. Furthermore, the weights pretrained on ImageNet may not provide a significant advantage for the specific geometric patterns required for traffic sign recognition.

Logistic Regression provides robust performance with exceptionally fast training and inference times. However, its high-dimensional feature space limits its interpretability compared to the CNN-based heatmaps. As previously noted, enhancing its transparency would require mapping HOG feature importance back to their spatial coordinates in the input image. Nevertheless, it remains a strong baseline.

Finally, both the shallow CNN and ResNet-18 offer impressive results. While ResNet-18 is the most capable model, its complexity and reduced transparency mean it is not always the optimal choice. In scenarios where interpretability is critical and a marginal decrease in accuracy is acceptable, a shallow CNN may serve as a superior alternative for safety-critical applications.

5 Discussion

This project explored a comprehensive explainability use case for Traffic Sign Recognition, evaluating the performance of four distinct architectures and using XAI techniques to obtain deeper insights into their decision-making processes. The study compared two baseline architectures (Logistic Regression over HOG features and a shallow CNN) against two sophisticated models, MobileNet-V2 and ResNet-18. While the baselines were trained from scratch on the GTSRB dataset, the complex architectures used transfer learning from ImageNet followed by fine-tuning. Regarding explainability, local insights were generated using Grad-CAM and Integrated Gradients, while global explanations were derived by aggregating individual Grad-CAM heatmaps. Both methods were validated using an Occlusion Sensitivity approach.

The results demonstrate the necessity of applying multiple XAI methods simultaneously, as their effectiveness varies across different scenarios and models. This study highlights the importance of a robust evaluation framework, such as Occlusion Sensitivity, to determine whether specific explanations are faithful to the model’s internal logic. Specifically, while Integrated Gradients often provide highly localized attributions, they occasionally diverge from the regions emphasized by Grad-CAM and Occlusion. Furthermore, the selection of the target layer for gradient-based methods was critical: the depth of the observed layer can affect whether a technique provides a clear semantic interpretation or an unusable representation.

Finally, the project repository contains a Jupyter notebook with all the code and experiments carried out. This allows users to conveniently replicate the experiments and generate custom explanations across the various models and XAI methods discussed.

References

- [1] M. Abdeldayem. Gtsrb - german traffic sign recognition benchmark. <https://www.kaggle.com/datasets/meowmeowmeowmeowmeow/gtsrb-german-traffic-sign>, 2018.
- [2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.
- [3] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. Mobilenetv2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4510–4520, 2018.

Appendix A Training Plots

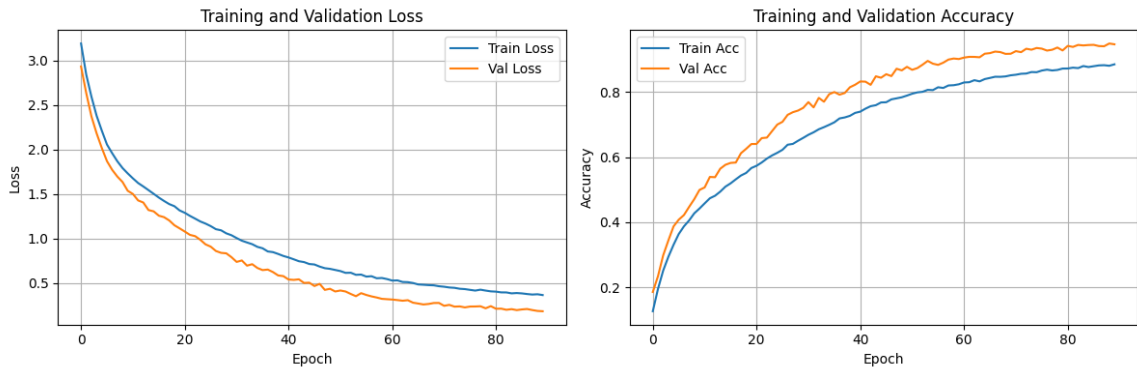


Figure 8: Shallow CNN training curves.

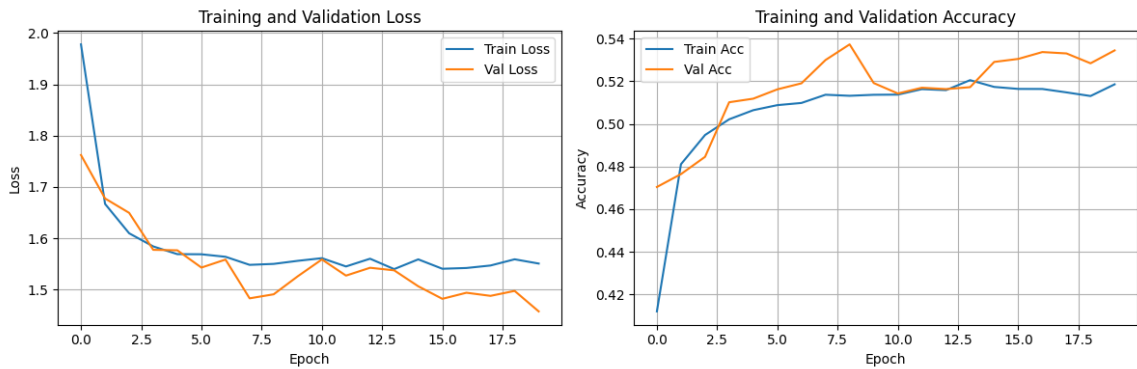


Figure 9: MobileNet-V2 phase 1 training curves.

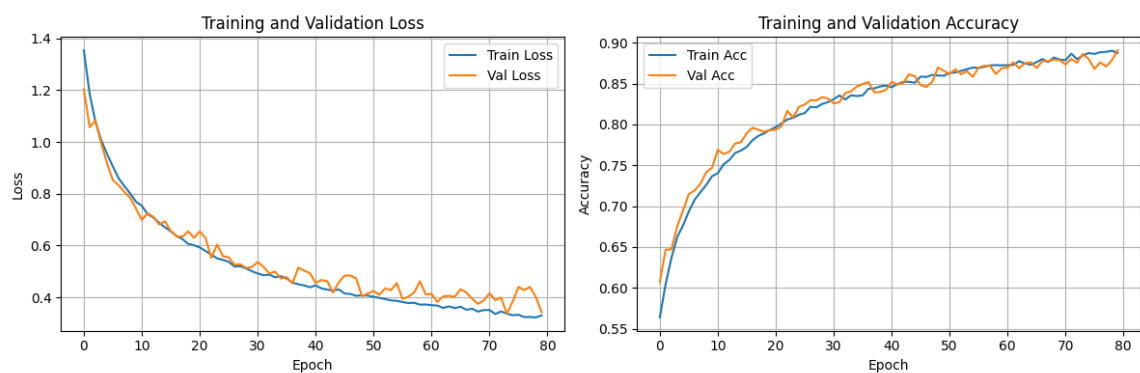


Figure 10: MobileNet-V2 phase 2 training curves.

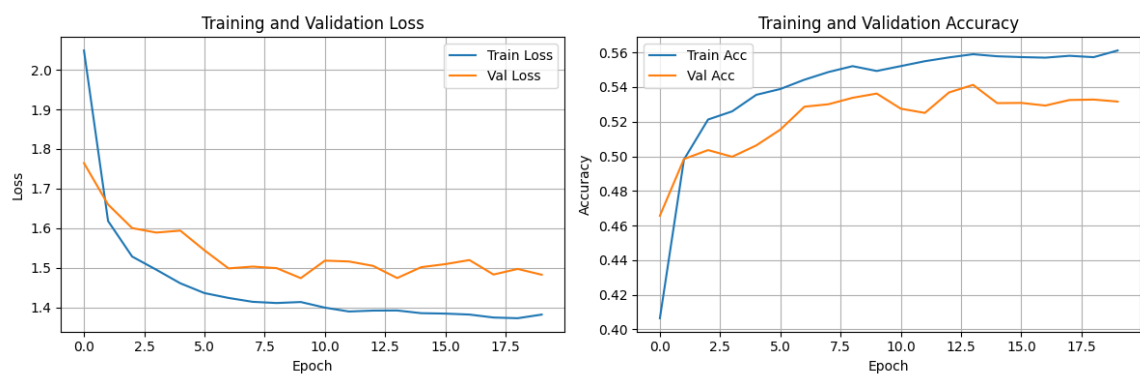


Figure 11: ResNet-18 phase 1 training curves.

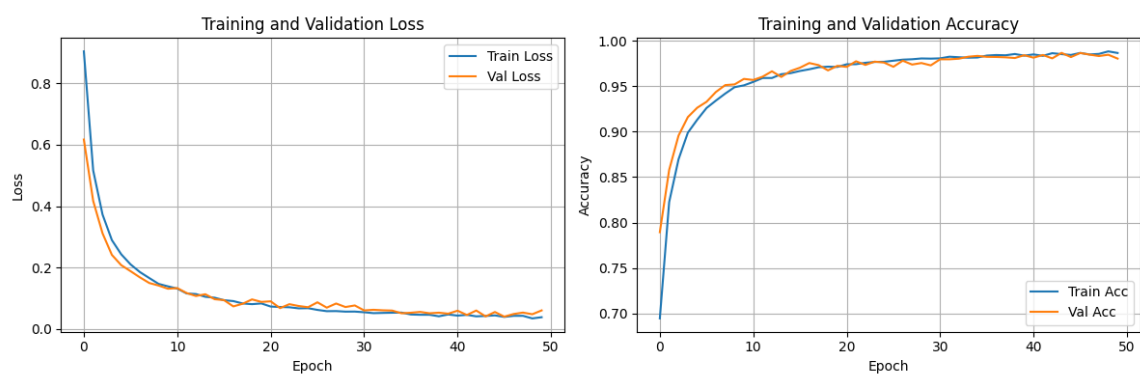


Figure 12: ResNet-18 phase 2 training curves.

Appendix B Further Explanations

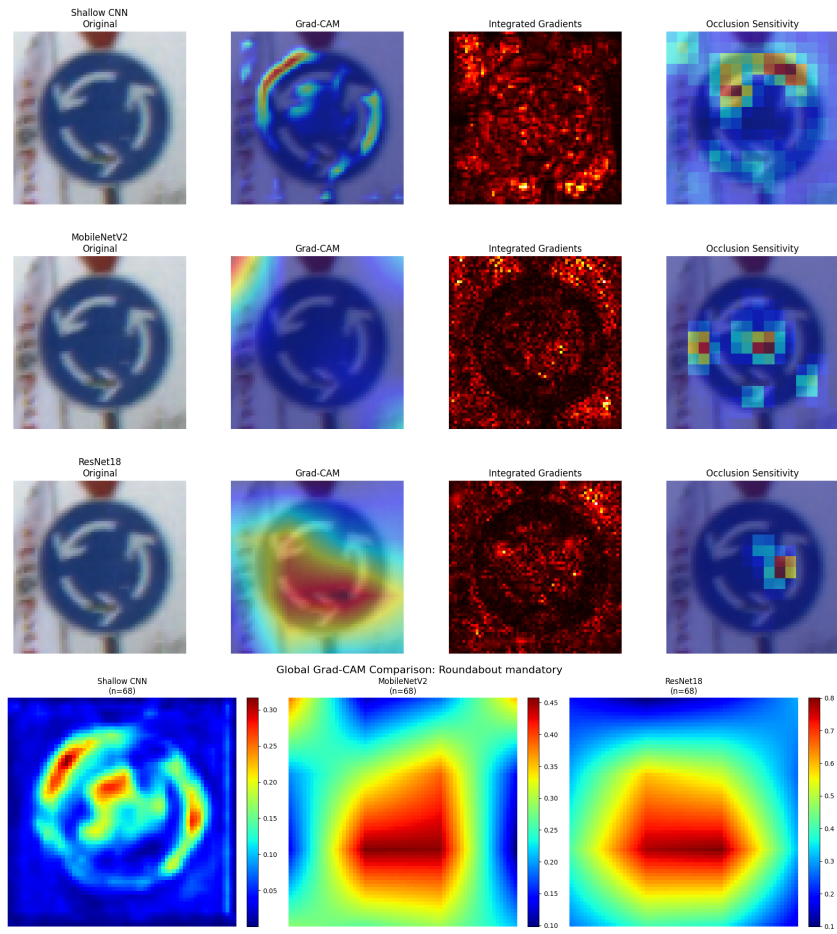


Figure 13: Local and global roundabout sign shows clearer visualizations from shallow CNN.

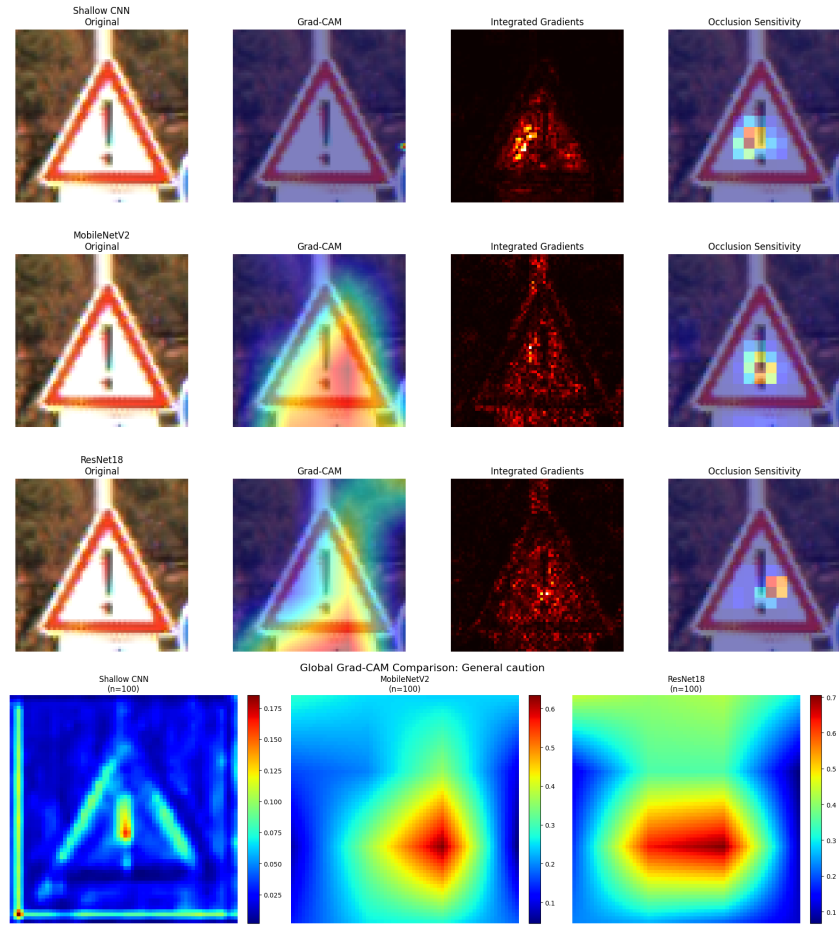


Figure 14: Local and global caution sign shows clearer visualizations from shallow CNN.